

GRAD: Gatekeeper & Reflex for Alzheimer's Disease

THE PATIENTS

Training Set
(n=320, ADNI)

Validation: A4 + Learn
n=1644

STAGE 1: GATEKEEPER

Univariate phospho-tau 217
Logistic Regression

Apply Probability Thresholds

$P < 0.25$, Amyloid Neg.
(n=98, 30.6%)

$0.25 \leq P \leq 0.75$, Gray Zone
(n=142, 44.4%)

$P > 0.75$, Amyloid Positive
(n=80, 25.0%)



STAGE 2: REFLEX

AUC: 0.755 on gray-zone subset

Random Forest Classifier

Multi-Marker Panel

GFAP p-tau 217 APOE ε4 Aβ42/40 Age

Classified Negative
-

Uncertain Cases
Refer to Amyloid-PET Scan

Classified Positive
+



MRI Enhancement:
Add +0.025 AUC to GRAD

GRAD Resolved: 87% of cases
Overall AUC: 0.867
Accuracy: 80.3%

PET Confirmation: 13% of cases